교육부 Ministry of Education

NRF 한국연구재단 National Research Foundation of Korea

공유교과목 [딥러닝]

Vector and Matrix

COSS Convergence and Open Sharing System 인공지능

전남대학교 CHONNAM NATIONAL UNIVERSITY

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서울과학기술대학교 SEOUL NATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY

영진전문대학교 YEUNGJIN UNIVERSITY

Scalar, Vector, Matrix and Tensor

Scalar, Vector, Matrix and Tensor

- Scalar : numbers without dimension

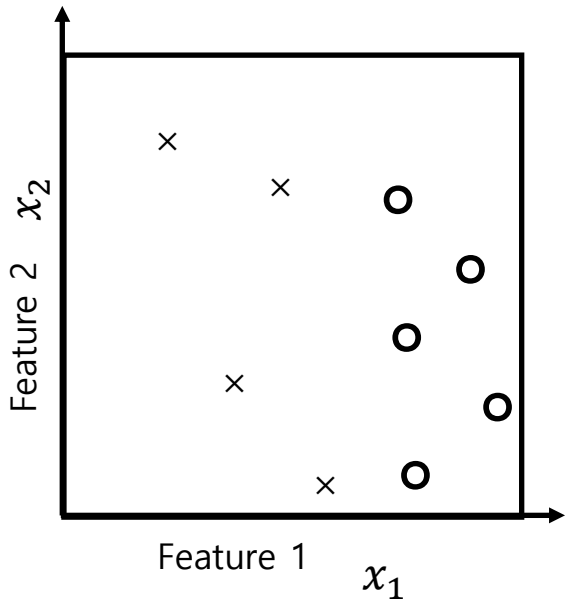
  - A scalar is 0-dimensional tensor.
  - Usually give scalars lower-case names written in italic.

    $s \in \mathbb{R}$

- natural number ($n \in \mathbb{N}$), real number ($s \in \mathbb{R}$) → n and s is a real-valued scalar

Scalar, Vector, Matrix and Tensor

- Vector: set of n numbers (n dimension vector)



- Example: positioning two data, which has separated features, in the "coordinate plane"

$$\vec{x} = (x_1, x_2)$$

$$\vec{x} = (x_1, x_2, x_3)$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Vector is 1-dimension tensor

Scalar, Vector, Matrix and Tensor

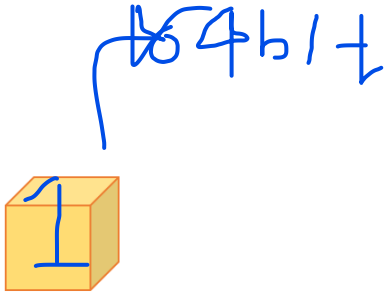
- Matrix: 2-dimension array of numbers

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}$$

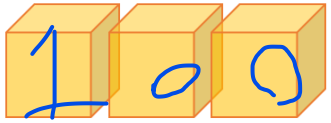
- $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a matrix of size $m \times n$, where $a_{i,j}$ is the element at the ^{i}th row and the ^{j}th column
- Matrix is 2-dimension tensor

Scalar, Vector, Matrix and Tensor

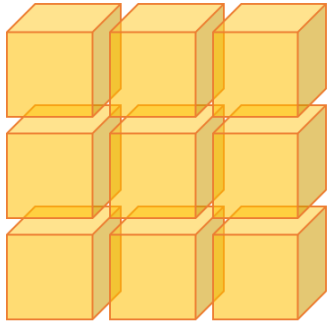
- Tensor: it is a n-dimension array, tensors are generalization of matrices



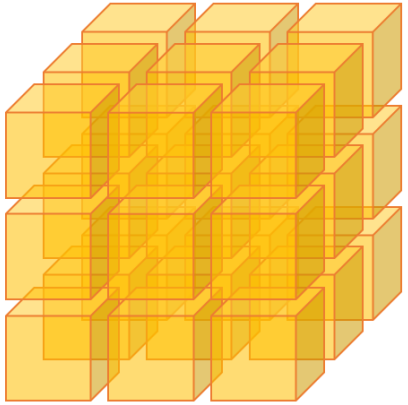
0-dimension tensor
(scalar)



1-dimension tensor
(vector)



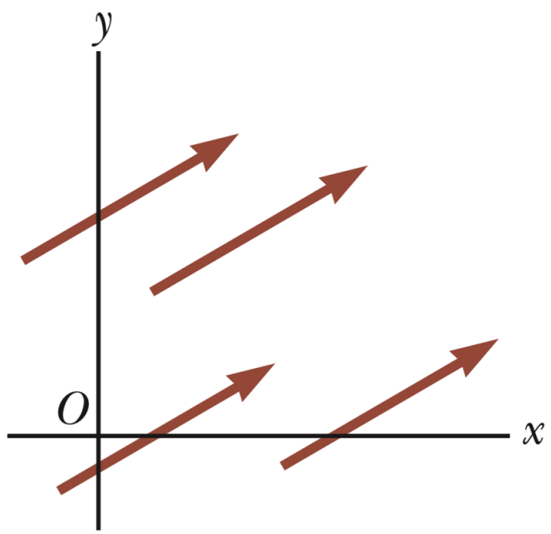
2-dimension tensor
(matrix)



3-dimension tensor

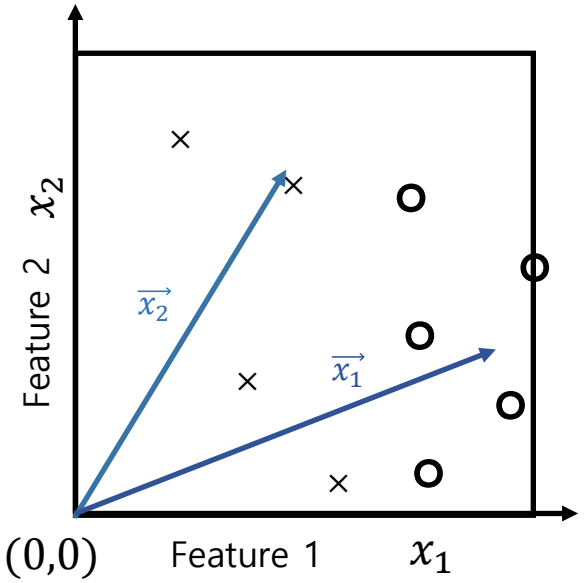
Vector Operation

Vector Operation



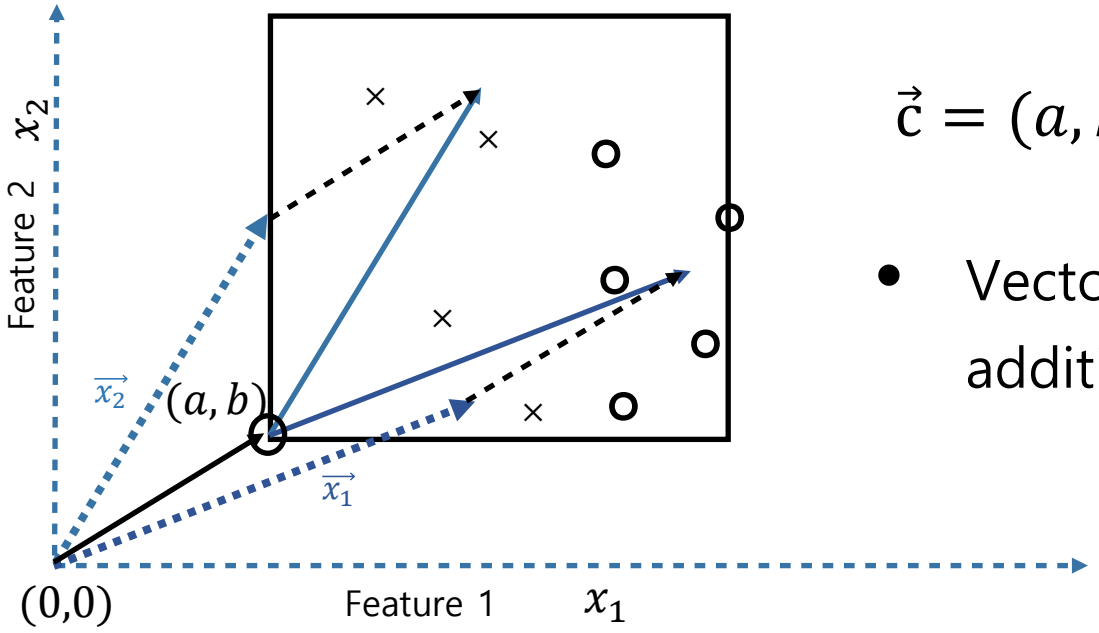
- If two vector has **"same size"** and **"same direction"**, it is a same vector

$$\overrightarrow{x_1} = (5, 2) \qquad \overrightarrow{x_2} = (2, 5)$$



- The number elements are used for analysis, the relative position of the data is important in coordinate component

Vector Operation – Positional data



$$\vec{c} = (a, b) \rightarrow \mathbf{c} = \begin{bmatrix} a \\ b \end{bmatrix}$$

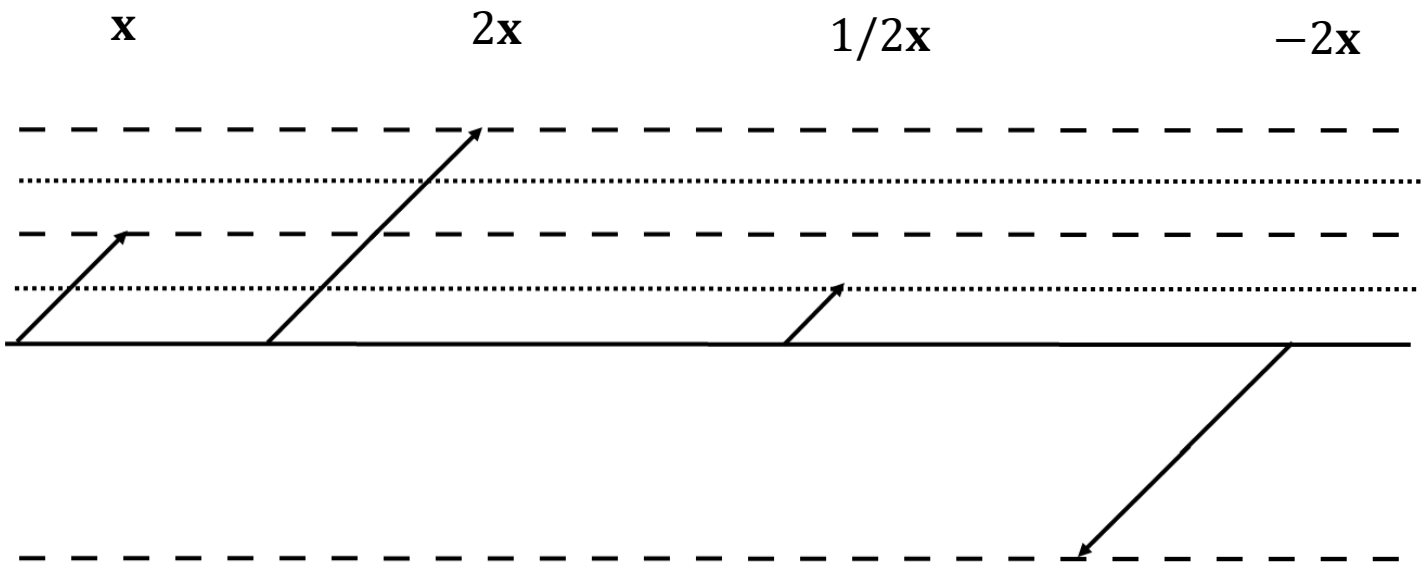
- Vector Operation:
  addition and subtraction

Vector – scalar multiplication (scaling)

- Vector $\mathbf{x}$ can be multiplied by the number s to get $s\mathbf{x}$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow s\mathbf{x} = \begin{bmatrix} sx_1 \\ sx_2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad -2\mathbf{x} = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$$



Vector Operation - addition

- Consider two vectors $\mathbf{u}$ and $\mathbf{v}$, if they have same size (shape), we can add them

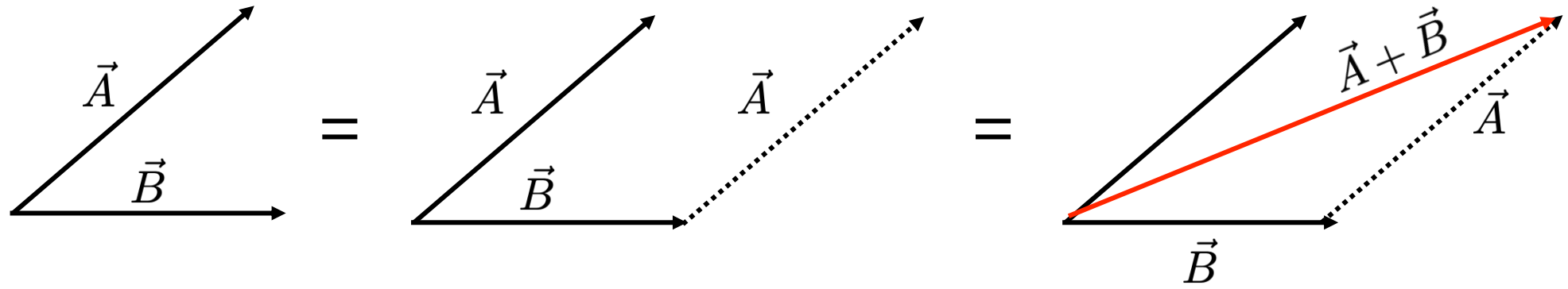
$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \longrightarrow \boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

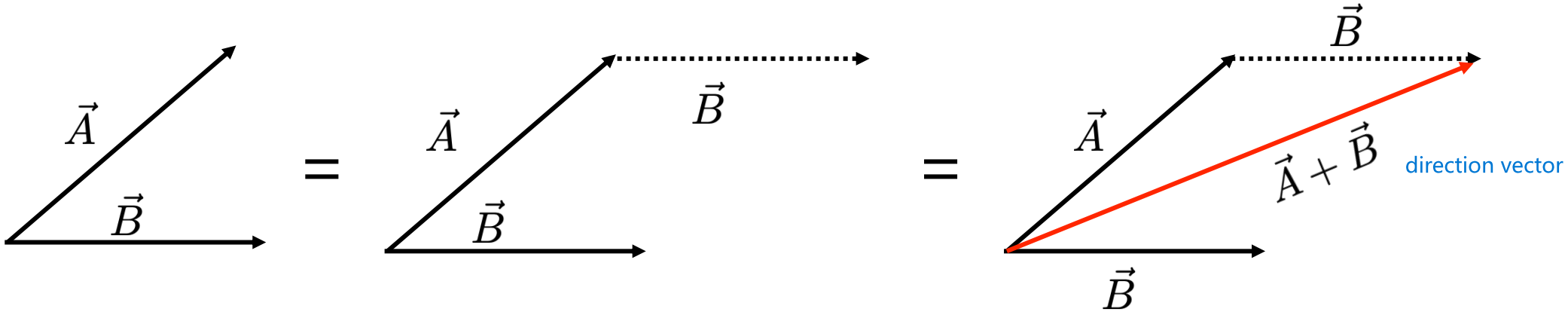
- Consider two vectors $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Then, vector addition between them given by

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

- Subtraction is defined in the same way. Given two vectors $\mathbf{u}$ and $\mathbf{v}$, the vector subtraction $\mathbf{u} - \mathbf{v}$ is equal to perform a scalar multiplication to make $-\mathbf{v}$ and then to add $\mathbf{u}$ and $-\mathbf{v}$.

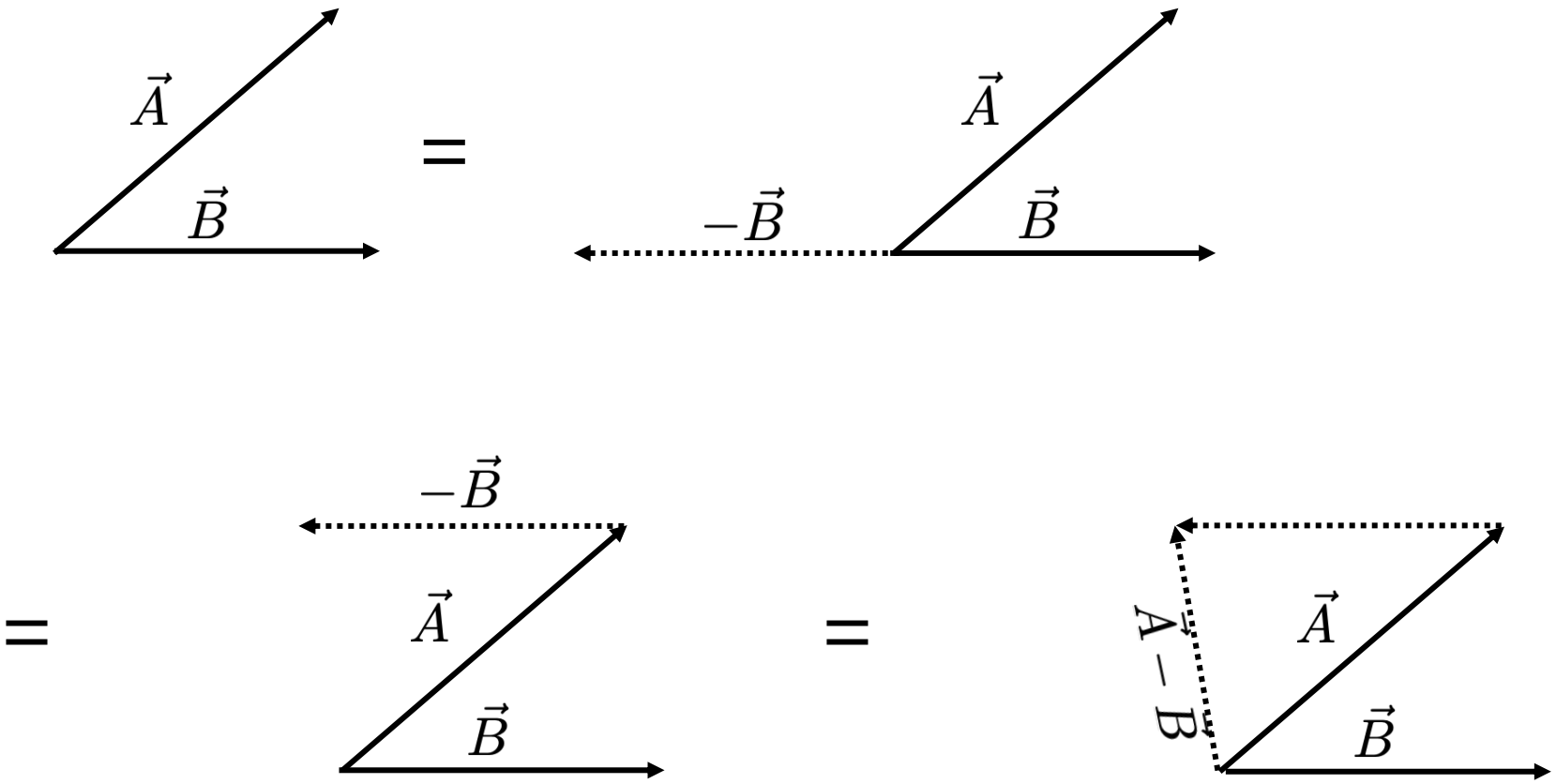
Vector Operation - addition





Vector Operation - subtraction

$$\vec{\mathbf{A}} - \vec{\mathbf{B}} = \vec{\mathbf{A}} + (-\vec{\mathbf{B}})$$



Vector Operation - multiplication

- The dot product (inner product) of two vectors $\mathbf{x}$ and $\mathbf{y}$ is the number $\mathbf{x} \cdot \mathbf{y}$ (or $\mathbf{x}^{\mathbf{T}}\mathbf{y}$)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \longrightarrow \mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2$$

- Another notation gives us an intuition about the dot product of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$

$$\mathbf{x}^T = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \longrightarrow \mathbf{x}^T\mathbf{y} = x_1 y_1 + x_2 y_2$$

$$(1 \times 2) \qquad (2 \times 1) \qquad\qquad (1 \times 1)$$

Vector Operation – transpose matrix

- Transpose : a matrix caused from flipping the rows and columns

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$,its transpose $\mathbf{A}^T \in \mathbb{R}^{n \times n}$,is defined as $(\mathbf{A}^T)_{i,j} = \mathbf{A}_{j,i}$

$$\mathbf{A} = \begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 1 & 9 \end{bmatrix} \qquad \mathbf{A}^T = \begin{bmatrix} 2 & 3 & 1 \\ 7 & 8 & 9 \end{bmatrix}$$

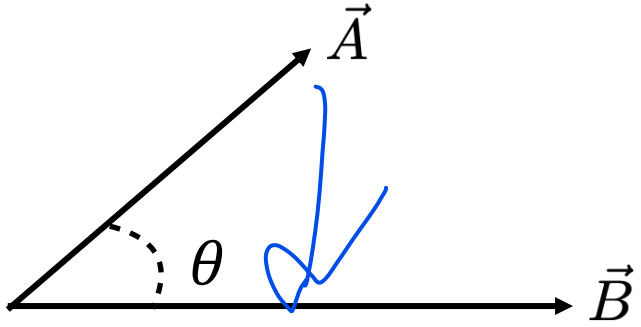
A vector has one column. The transpose is one row matrix

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad \mathbf{x}^T = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

The transpose of a scalar is itself

$$s = s^T$$

Vector vs. Matrix – dot product



$$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta$$

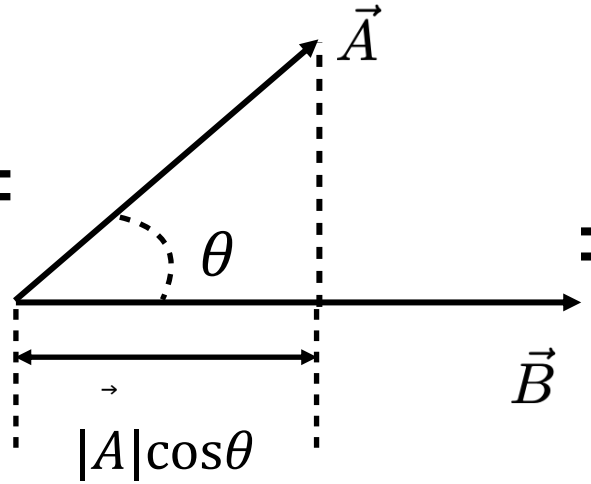
$$|\vec{A}| = \sqrt{A_1^2 + A_2^2 + \cdots + A_n^2}$$

$$|\vec{B}| = \sqrt{B_1^2 + B_2^2 + \cdots + B_n^2}$$

cosine simularity

A 와 B 의

$$\vec{A} \cdot \vec{B} =$$



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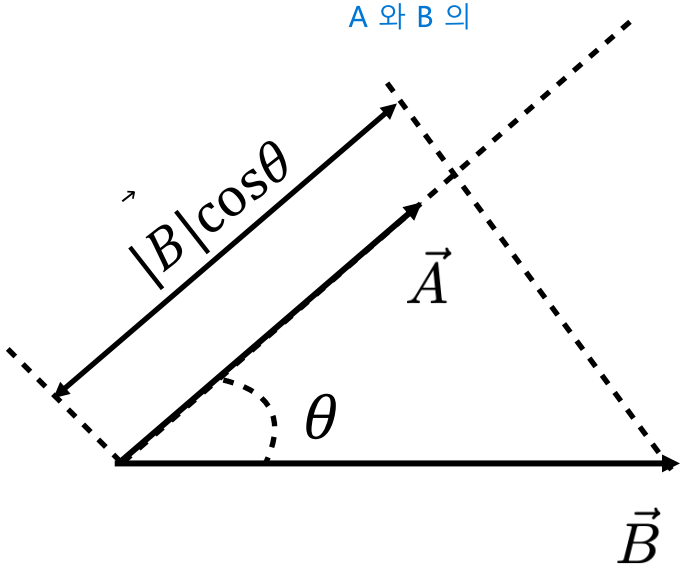
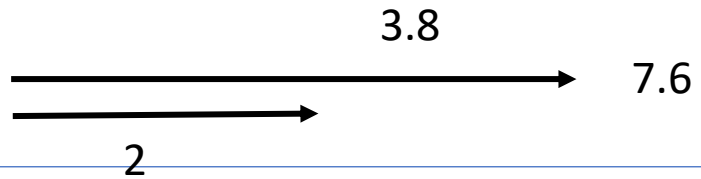
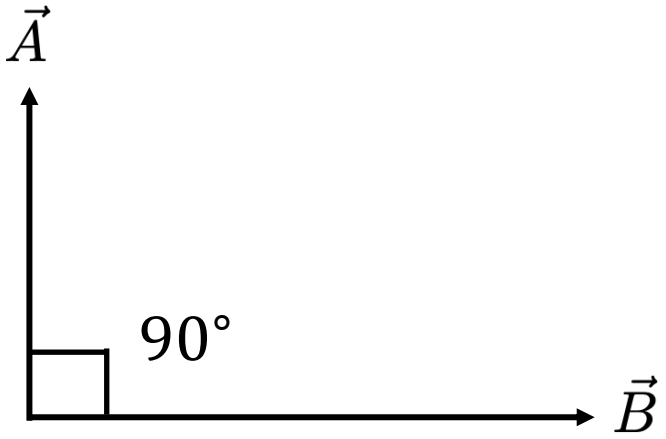




Fit the calculation criteria on one axis

$$|\vec{A}| \cos\boldsymbol{\theta} \cdot |\vec{B}|$$

Vector vs. Matrix – dot product



$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos 90° = 0$$

$\vec{A} \cdot \vec{A} = |\vec{A}||\vec{A}| = |\vec{A}|^2$ (Vector dot product, vector size)

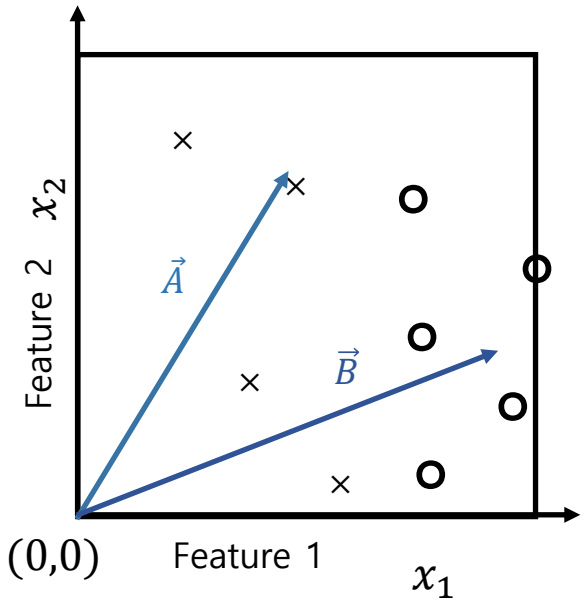
$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ (Commutative Law)

$(\vec{A} + \vec{B}) \cdot \vec{C} = \vec{A} \cdot \vec{C} + \vec{B} \cdot \vec{C}$ (Distributive Law)

$(a\vec{A}) \cdot \vec{B} = a\,(\vec{A} \cdot \vec{B}) = \vec{A} \cdot (a\vec{B})$ (Associative Law)

Vector vs. Matrix – dot product

- Dot product vs. matrix multiplication with transpose matrix



그, 두 기저 벡터

$$\vec{A} = (A_1, A_2) = A_1\hat{\imath} + A_2\hat{\jmath}$$

Axis reference change

$$\vec{B} = (B_1, B_2) = B_1\hat{\imath} + B_2\hat{\jmath}$$

$$\vec{A}\cdot\vec{B} = (A_1\hat{\imath} + A_2\hat{\jmath})\cdot(B_1\hat{\imath} + B_2\hat{\jmath})$$

$$= A_1B_1\,\hat{\imath}\cdot\hat{\imath} + A_1B_2\,\hat{\imath}\cdot\hat{\jmath} + A_2B_1\,\hat{\jmath}\cdot\hat{\imath} + A_2B_2\,\hat{\jmath}\cdot\hat{\jmath}$$

$$= A_1B_1 + A_2B_2$$

$(\hat{\imath}\cdot\hat{\imath} = \hat{\jmath}\cdot\hat{\jmath} = 1,\ \hat{\imath}\cdot\hat{\jmath} = \hat{\jmath}\cdot\hat{\imath} = 0)$

$\vec{A}$ or $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $\quad \vec{B}$ or $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ $\qquad \mathbf{x}\cdot\mathbf{y} = x_1y_1 + x_2y_2$

Vector vs. Matrix – dot product

Dot product of vectors can be changed with **matrix multiplication**

The dot product (inner product) of two vectors $\mathbf{x}$ and $\mathbf{y}$ is the number $\mathbf{x} \cdot \mathbf{y}$ (or $\mathbf{x}^{\mathrm{T}}\mathbf{y}$)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \qquad \longrightarrow \qquad \mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2$$

Another notation give us an intuition about the dot product of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$

Matrix multiplication with transpose operation

$$\mathbf{x}^T = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \qquad \longrightarrow \qquad \mathbf{x}^T\mathbf{y} = x_1 y_1 + x_2 y_2$$

$$(1 \times 2) \qquad (2 \times 1) \qquad\qquad (1 \times 1)$$

Vector operation manipulated with matrix operation

Vector vs. Matrix

- The product of a matrix and a vector is defined in the same way

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \longrightarrow \mathbf{Ax} = \begin{bmatrix} a_{1,1}x_1 + a_{1,2}x_2 + a_{1,3}x_3 \\ a_{2,1}x_1 + a_{2,2}x_2 + a_{2,3}x_3 \end{bmatrix}$$

$$(2 \times 3) \qquad (3 \times 1) \qquad (2 \times 1)$$

- The product of two matrices … (**all vector operations are manipulated with matrix**)

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} \longrightarrow \mathbf{AB} = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + a_{1,3}b_{3,2} \\ a_{2,1}b_{1,1} + a_{2,2}b_{2,1} + a_{2,3}b_{3,1} & a_{2,1}b_{1,2} + a_{2,2}b_{2,2} + a_{2,3}b_{3,2} \end{bmatrix}$$

$$(2 \times 3) \qquad (3 \times 2) \qquad (2 \times 2)$$

Vector Operation – Properties I

- Addition Laws
  - $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ (commutative law)
  - $c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}$ (distributive law)
  - $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$ (associative law)

- Multiplication Laws
  - $\mathbf{AB} \neq \mathbf{BA}$ (commutative law is usually broken)
  - $\mathbf{C}(\mathbf{A} + \mathbf{B}) = \mathbf{CA} + \mathbf{CB}$ (distributive law from the left)
  - $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$ (distributive law from the right)
  - $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$ (associative law for $\mathbf{ABC}$)(parentheses not needed)

Vector Operation – Properties II

- Transpose matrix
  - $(\mathbf{A}^T)^T = \mathbf{A}$
  - $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
  - $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$
  - $(c\mathbf{A})^T = c\mathbf{A}^T$

Vector Norm

- Norm is a size of vector (Norm)

L^p norm: $\| \vec{x} \|_p = (\sum |x_i|^p)^{1/p} \quad (p \geq 1)$

- It is a measurement for the length of a vector

- A function $f: \mathbb{R}^n \to \mathbb{R}$ that satisfies 4 properties:
  - For all $\mathbf{x} \in \mathbb{R}^n$,$f(\mathbf{x}) \geq 0$ (non-negativity)
  - $f(\mathbf{x}) = 0$ if and only if $\boldsymbol{x} = \mathbf{0}$ (definiteness)
  - For all $\mathbf{x} \in \mathbb{R}^n$, $c \in \mathbb{R}$, $f(c\mathbf{x}) = |c| f(\mathbf{x})$ (absolute value homogeneity)
  - For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$ (triangle inequality)

    *One of the essential conditions that must be satisfied for us to use the function f(x)f(x)f(x) as a **norm** or a **metric function**.

- where $\mathbf{0}$ is a zero vector whose all elements are zero

Example 1: Euclidean Norm (Vector Case)

The Euclidean norm is defined as:

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2}$$

Now, let's verify whether the triangle inequality:

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

holds for two vectors $\mathbf{x}$ and $\mathbf{y}$.

Step 1: Compute the norm of each vector

Given the vectors $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (-1, 2)$:

$$\|\mathbf{x}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\|\mathbf{y}\| = \sqrt{(-1)^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$$

Step 2: Compute the sum of the vectors

$$\mathbf{x} + \mathbf{y} = (3 + (-1), 4 + 2) = (2, 6)$$

Step 3: Compute the norm of the resulting vector

$$\|\mathbf{x} + \mathbf{y}\| = \sqrt{2^2 + 6^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$$

Step 4: Verify the triangle inequality

$$2\sqrt{10} \leq 5 + \sqrt{5}$$

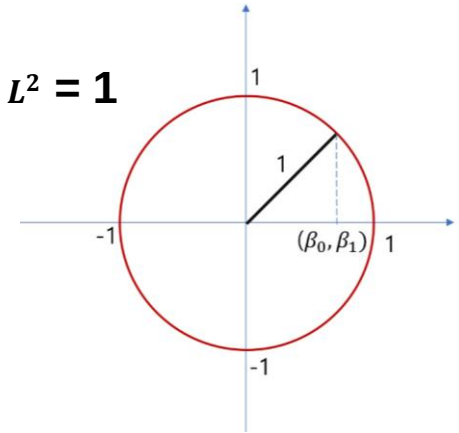
Since this inequality holds, we confirm that the triangle inequality is satisfied.

Vector Norm – L^1 Norm, L^2 Norm

- $\boldsymbol{L^2}$ norm (Euclidean norm) $= \sqrt{\vec{x} \cdot \vec{x}}$, square function $\boldsymbol{L^2}$ norm $= \vec{x} \cdot \vec{x}$

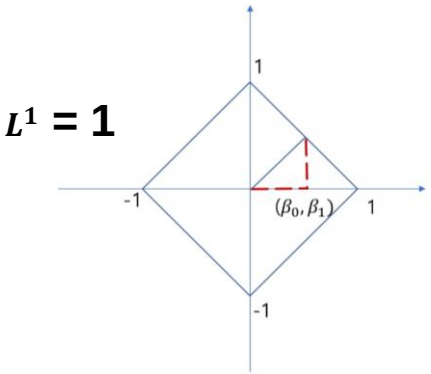
  - Simply square function $\boldsymbol{L^2}$ norm used for loss function

$$x = [1, 2, 3, 4, 5] \qquad \|x\|_2 = \sqrt{(|1|^2 + |2|^2 + |3|^2 + |4|^2 + |5|^2)}$$
$$= \sqrt{1 + 4 + 9 + 16 + 25}$$
$$= \sqrt{55}$$
$$= 7.4161$$



- $\boldsymbol{L^1}$ norm $\| \vec{x} \|_1 = \sum_i |x_i|$ or $\boldsymbol{L^\infty}$ norm $\| \vec{x} \|_\infty = \max_i |x_i|$

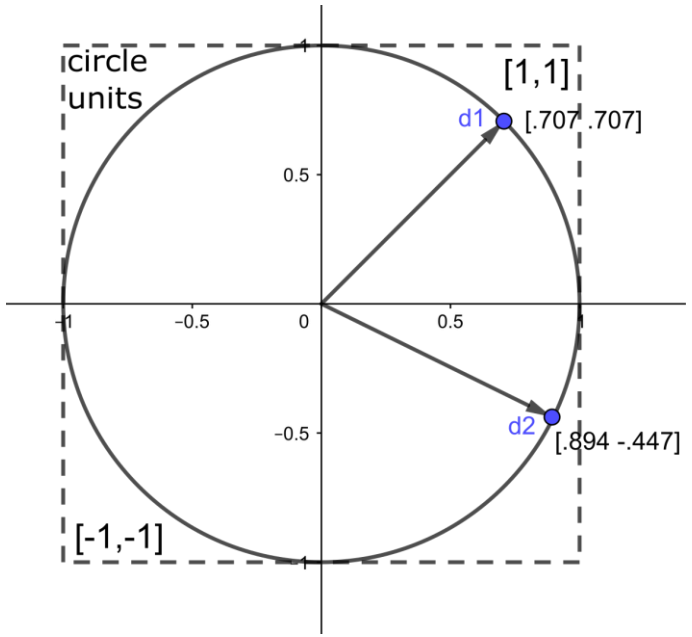
$$x = [1, 2, 3, 4, 5] \qquad \|x\|_1 = (|1| + |2| + |3| + |4| + |5|)$$
$$= 15$$



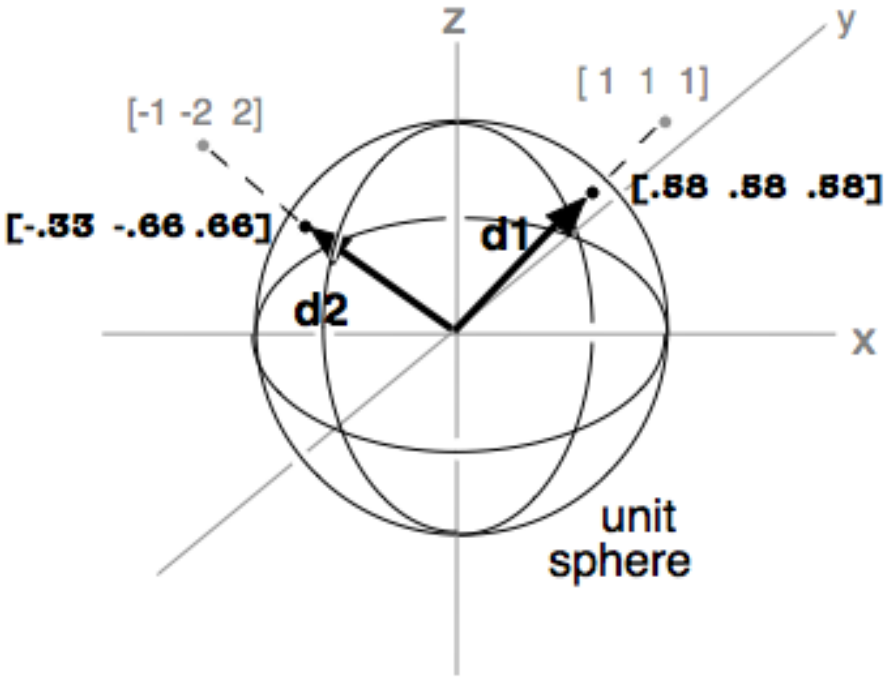
Unit Vector

Vector Norm – Unit vectors

- A unit vector is a vector with unit norm : $\|\boldsymbol{x}\|_2 = 1$



2-dimensional unit vectors



3-dimensional unit vectors

Unit vector vs. Basis vector

- Unit vector:
  - Has a magnitude of 1.
  - Used to represent direction.
  - Can be created by dividing any vector by its magnitude.

- Basis vector:
  - Part of a set of vectors that can represent any vector in a given vector space.
  - Requires linear independence within the set.
  - May or may not have a magnitude of 1.

- Example
  - In a 2D plane, the vectors (1, 0) and (0, 1) are considered basis vectors because any other point on the plane can be expressed as a combination of these two. However, they are also unit vectors because their magnitude is 1.

Vector – one-hot vectors

- One-hot vector has just one “1” element and the others are all “0”

[0. 0. 1. 0. 0. 0. 0. 0.] #Index 2, one-hot vector
[0. 0. 0. 0. 0. 1. 0. 0.] #Index 5, one-hot vector
[0. 1. 0. 0. 0. 0. 0. 0.] #Index 1, one-hot vector
[0. 0. 0. 0. 0. 0. 1. 0.] #Index 6, one-hot vector
[0. 0. 0. 1. 0. 0. 0. 0.] #Index 3, one-hot vector
[0. 0. 0. 0. 0. 0. 0. 1.] #Index 7, one-hot vector

c l a s s i f i c a t i o n

- In machine learning, one-hot-vectors are used to represent a sequence or index
  - In Classification, it is used to represent labels (select one)
  - In Natural Language Process, it is used to encode words (represent a word position)

Vector – identity matrix

An identity matrix $\mathbf{I}$ has diagonal elements a_{ii} of 1 and the other elements of 0.

They are identity matrices

$$[1] \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

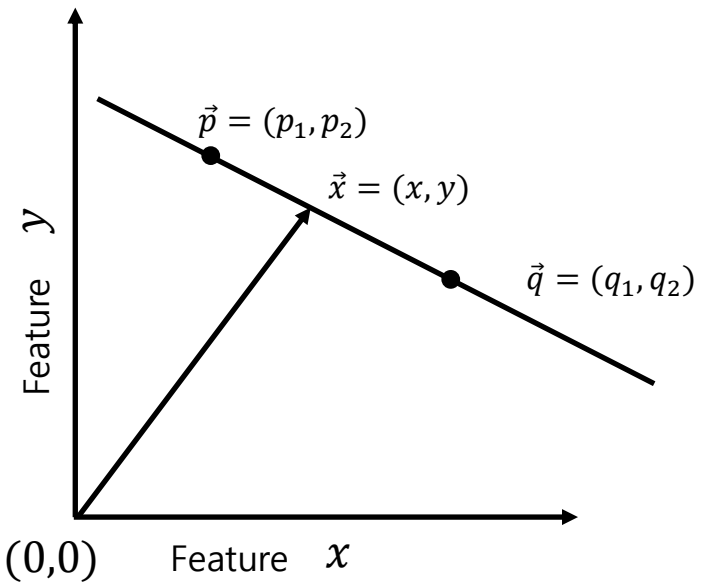
Like scalar 1, identity matrices are the identity of multiplication:

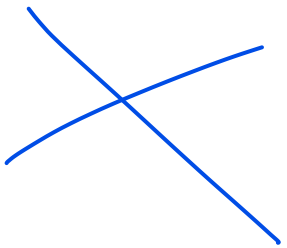
- $\mathbf{AI} = \mathbf{A}$
- $\mathbf{IA} = \mathbf{A}$
- $\mathbf{AA}^{-1} = \mathbf{I}$
- $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$

where $\mathbf{A}^{-1}$ is the inverse matrix of $\mathbf{A}$.

Vector – line

- A straight line passing through two points is displayed using a vector





$$\overrightarrow{px} \propto \overrightarrow{pq}$$

$$\overrightarrow{px} = \ell \overrightarrow{pq}$$

$$\overrightarrow{px} = \vec{x} - \vec{p} = (x - p_1, y - p_2)$$

$$\overrightarrow{pq} = \vec{q} - \vec{p} = (q_1 - p_1, q_2 - p_2)$$

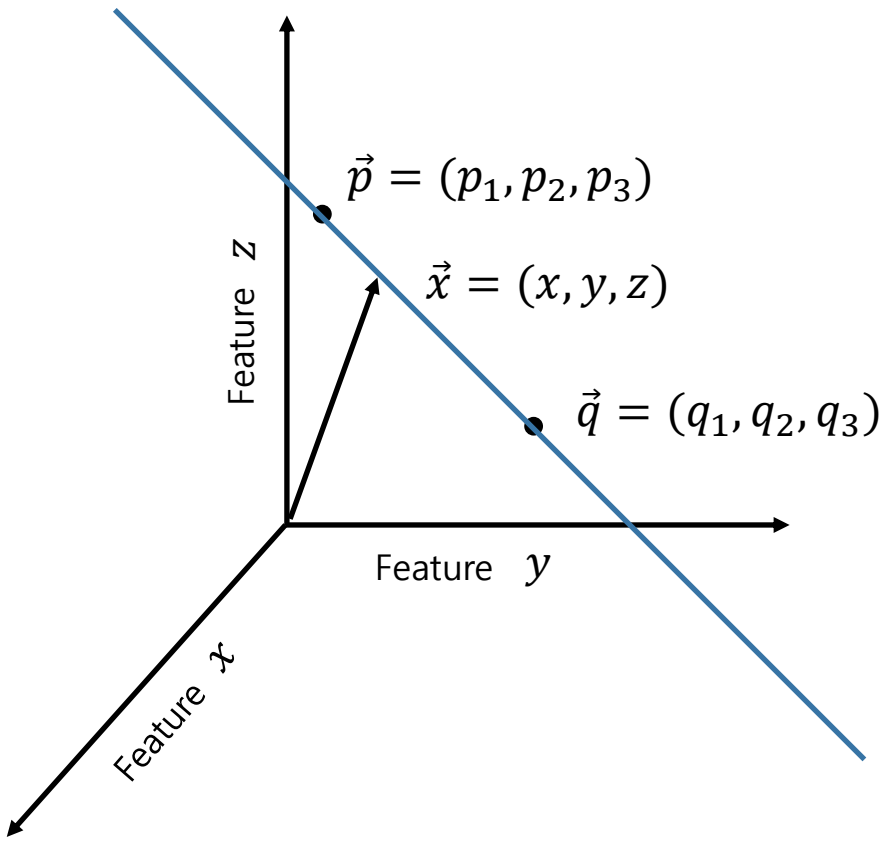
$$\overrightarrow{px} = \ell \overrightarrow{pq} \rightarrow \quad (x - p_1, y - p_2) = \ell(q_1 - p_1, q_2 - p_2)$$

$$\frac{x - p_1}{q_1 - p_1} = \frac{y - p_2}{q_2 - p_2} \quad \Rightarrow \quad y = \frac{q_2 - p_2}{q_1 - p_1}(x - p_1) + p_2$$

Vector Norm – n-dimension vector

- Display a straight line passing through two points using a vector:

  easy to expand to multiple dimensions



$$\overrightarrow{px} \propto \overrightarrow{pq}$$

$$\overrightarrow{px} = \vec{x} - \vec{p} = (x - p_1, y - p_2, z - p_3)$$

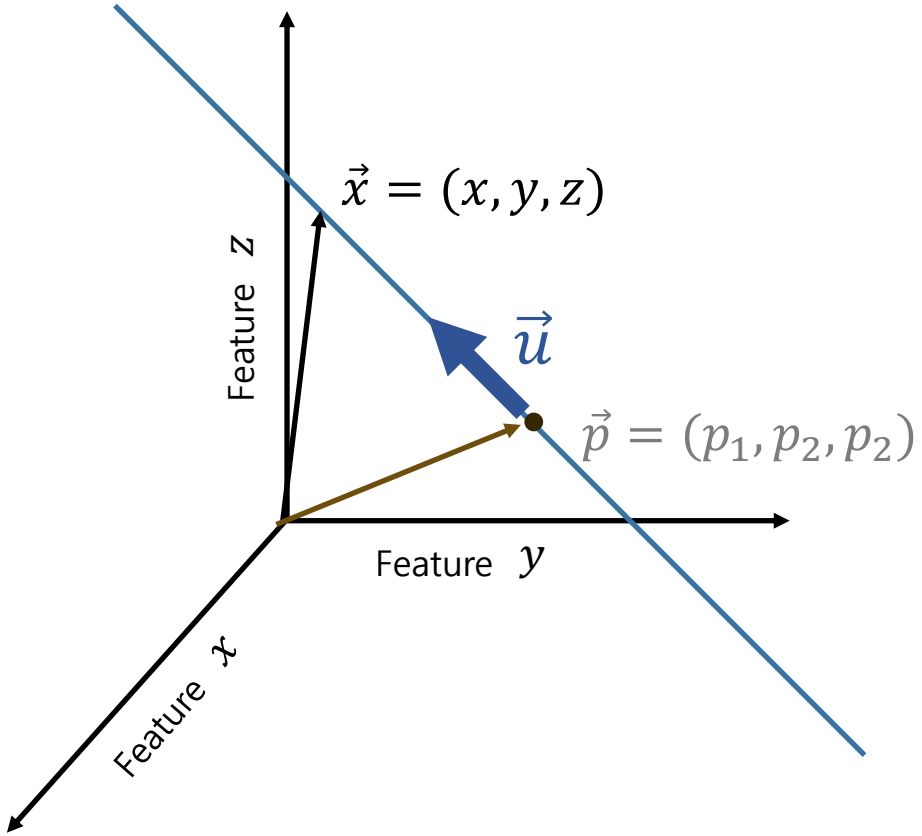
$$\overrightarrow{pq} = \vec{q} - \vec{p} = (q_1 - p_1, q_2 - p_2, q_3 - p_3)$$

$$\frac{x - p_1}{q_1 - p_1} = \frac{y - p_2}{q_2 - p_2} = \frac{z - p_3}{q_3 - p_3}$$

Vector – display with unit vector

- A straight line passing through the point $\vec{p}$and parallel to the vector $\vec{u}$:

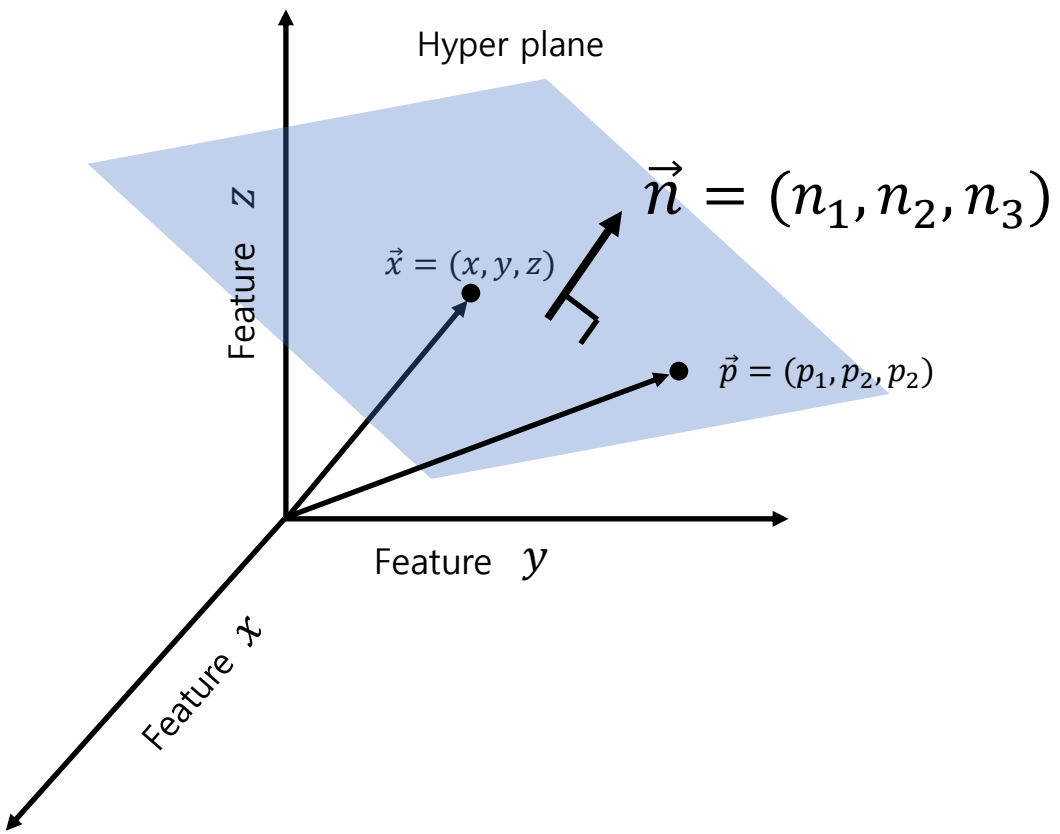
  simply expressed as the sum of the vectors



$$\vec{x} = \vec{p} + t\vec{u}$$

Vector Plane

- Equation of a plane: Equation of a plane using a normal vector, (a vector normal to the plane, normal vector)



$$\overrightarrow{px} = \vec{x} - \vec{p} = (x - p_1, y - p_2)$$

$$\overrightarrow{px} \perp \vec{n}$$

$$\overrightarrow{px} \cdot \vec{n} = 0$$

$$(\text{matrix } \overrightarrow{px} \cdot \vec{n}^T = 0)$$

$$\overrightarrow{px} = \vec{x} - \vec{p} = (x, y, z) - (p_1, p_2, p_3)$$

$$\overrightarrow{px} \cdot \vec{n} = (x - p_1, y - p_2, z - p_3) \cdot (n_1, n_2, n_3) = 0$$

$$n_1(x - p_1) + n_2(y - p_2) + n_3(z - p_3) = 0$$

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thank you

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